Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS) Sem-I Remedial April-June 2024

U61A1IP1: INTRODUCTION TO PROGRAMMING-I

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Describe the structure of a C program, detailing its components and how they contribute to the overall functionality of the program.	06	CO2	BTL2
2	Explain the significance of arithmetic, relational, logical, and assignment operators in C programming, and discuss their roles in creating efficient and effective code.	06	CO3	BTL2
3	Focus on looping statements such as WHILE, DO-WHILE, or FOR to solve a specific problem, provide valid syntax and proper example for each.	06	CO4	BTL4
4	Develop algorithms and draw a flowchart for basic human general activities, such as making a sandwich or making a cup of tea.	06	CO1	BTL3
5	Evaluate and implement different string handling functions in C programming.	06	CO5	BTL6
6	Explain the concepts of constants, variables, and data types in C programming, highlighting their roles in storing and managing data.	06	CO2	BTL2
7	Write a C program using the ternary (?:) operator to determine whether a given number is even or odd, demonstrating its compact syntax for conditional expressions.	06	CO3	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Compare and contrast the characteristics of primitive data types and derived data types in C programming, providing examples to illustrate your points.	06	CO2	BTL4
2	Describe the importance of the C programming language in the context of modern software development, highlighting its key features and advantages.	06	CO2	BTL2

3	Explain the purpose and components of a flowchart, highlighting its importance in visualizing the logical flow of a process or algorithm.	06	CO1	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the concept of decision making in C programming using the IF statement, discussing its syntax and basic usage.	06	CO4	BTL2
2	Compare and contrast the role and behavior of increment-decrement operators and conditional operators in C programming, providing examples to demonstrate their usage and differences.	06	CO3	BTL4
3	Explain the purpose and syntax of the switch statement in C programming, highlighting its use for multiple branching based on the value of an expression.	06	CO3	BTL2
Q.4	**Answer the following: (Any One)**	**06**		
1	Create an array to store the marks of students and initialize it with sample values, then perform calculations to find the total and percentage.	06	CO5	BTL6
2	Explain the concept of arrays in programming, focusing on their role in storing multiple elements of the same data type and the concept of dimensions in arrays.	06	CO5	BTL2

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Cyber Security) Sem-I Remedial April-June 2024

U61A3OS1 OPERATING SYSTEM SECURITY-1

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	What is Windows? Define the history of the Windows Operating System.	06	CO1	BTLI2/BTL4
2	Discuss the DIR command with all attributes.	06	CO2	BTL6
3	Justify FAT32 and NTFS in detail.	06	CO1	BTL3
4	Discuss any one Process scheduling algorithm.	06	CO4	BTL5
5	Classify the following Privacy-protecting option. Camera, Call History, Microphone	06	CO3	BTL3
6	Discuss System Call with its type.	06	CO4	BTL4
7	Describe Paging with an example.	06	CO5	BTL6
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	What is Operating Systems? Classify its types.	06	CO1	BTLI2/BTL3
2	Compare DOS and Windows.	06	CO2	BTL5
3	**Discuss below commands** getmac, ipconfig, tracert	06	CO2	BTL2
Q.3	**Answer the following: (Any Two)**	**12**		
1	What is Process? Classify Process State Diagram.	06	C04	BTLI2/BTL4
2	Discuss the Family option in detail.	06	CO3	BTL5
3	Discuss the Firewall in detail	06	CO3	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	What is Deadlock? Discuss Deadlock characterization.	06	CO5	BTLI2/BTL3
2	Define Logical and Physical Address Space.	06	CO5	BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Cyber Security) Sem-I Remedial April-June 2024

U61A4FIS: Fundamental of Information Security

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Define Computer Networks. Discuss types of Computer Networks.	06	CO1	BTL1
2	What is CIA triad in information security?	06	CO2	BTL2
3	Differentiate between DOS and DDOS attack with example.	06	CO3	BTL4
4	Discuss different phishing attacks.	06	CO3	BTL3
5	Differentiate between plain vs cipher text & encryption vs. Decryption	06	CO4	BTL4
6	Justify the importance of physical security.	06	CO5	BTL5
7	Define VPN. Discuss the role of VPN.	06	CO5	BTL6
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	What is information security? Discuss the importance of information security.	06	CO1	BTL1
2	Differentiate between TPS, MIS and DSS.	06	CO1	BTL4
3	Discuss Email security attacks and counter measures.	06	CO3	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	How is symmetric key being different than asymmetric key cryptography?	06	CO4	BTL4
2	Explain various types of authentication methods.	06	CO5	BTL2
3	Define wireless network. Justify the need of authentication in wireless network.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Deliberate the role of firewall in network security.	06	CO5	BTL3
2	How are IPS and IDS helpful to prevent attacks?	06	CO5	BTL6

**** End of the Paper ****

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GANPAT UNIVERSITY

B.Sc.IT(CS)– Semester -I

Remedial - April-June 2024

U61A5FM: FUNDAMENTAL OF MATHEMATICS

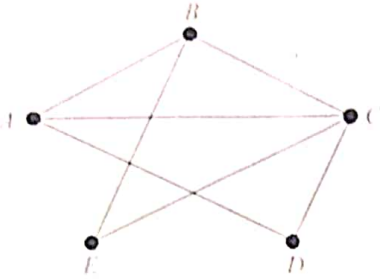
Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	Let $U = \{1,2,3,4,5,6,7,8\}$ be the Universal set and let $A = \{1,3,5,6\}$, $B = \{2,4,6,7\}$, $C = \{2,3,7,8\}$ Find (a) $B \cup C$ (b) $A \cap C$ (c) A/C (d) B/A (e) $B \oplus C$	06	CO1	BTL1, BTL2, BTL6
2	Each student in Liberal Arts at some college has a mathematics requirement A and a science requirement B. A poll of 150 sophomore students shows that: 70 completed A, 50 completed B, 21 completed both A and B. Use a Venn diagram to find the number of students who have completed: (a) At least one of A and B; (b) exactly one of A or B; (c) neither A nor B.	06	CO2	BTL4 & BTL5 & BTL6
3	Define square matrix.Also prove that A.(B+C)=A.B+A.C where $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 2 & 3\\ 1 & -2 & 2\end{bmatrix} \& B=\begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 1\\ 2 & 2 & 3\end{bmatrix} \& C=\begin{bmatrix}1 & 0 & 2\\ 2 & 4 & 0\\ -1 & 2 & 3\end{bmatrix}$	06	CO3	BTL1, BTL2, BTL6
4	Define the following function with example. (a) Injective function (B)Surjective function (c) In to function (d) Identity function (e) Equal function (f) Constant function	06	CO2	BTL2 &BTL3

5	Define Inverse Function. Also (a) If f:N $\rightarrow$N and f(x)=x+1. Does f^{-1} exist ? why? (b) If f:R$\rightarrow$R and f(x)=2x-3. Then find $f^{-1}(x)$ & $f^{-1}(9)$	06	CO4	BTL2&BTL 3&BTL5
6	Consider the graph G  (a)Describe G formally, that is, find the set V (G) of vertices of G and the set E(G) of edges of G. (b) Find the degree of each vertex and verify the sum of degree is twice the number of edges.	06	CO3	BTL3, BTL4, BTL5
7	Find the number n of distinct permutations that can be formed from all the letters of each word: (a) LEADER; (b) UNUSUAL; (c) RUMOUR.	06	CO2	BTL3, BTL4, BTL6

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GANPAT UNIVERSITY

B.Sc.IT (Cyber Security) Sem-I Remedial Theory April-June 2024

U61B6CS1: LANGUAGE & COMMUNICATION SKILLS-I

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy
1	Discuss the function of conjunctions in sentence construction. Provide examples of common conjunctions and their usage.	06	CO1	BTL1,4
2	Discuss the role of electronic and visual communication in today's digital age.	06	CO2	BTL2,6
3	Identify common barriers to effective listening and suggest strategies for overcoming them.	06	CO3	BTL2,5
4	Outline the techniques for improving reading comprehension, including active reading, previewing, annotating, and summarizing.	06	CO4	BTL2,3
5	Discuss the role of feedback and self-reflection in the process of honing oral communication abilities.	06	CO5	BTL1,2
6	Explain the equation **E=A+I** in terms of letter writing.	06	CO4	BTL2,6
7	Explain the concepts of coherence and cohesion in writing and their significance in conveying ideas clearly.	06	CO2	BTL1,3

Section – II

		Marks	CO	Bloom Taxono
Q-2	**Answer the following (Any Two)**	30		
1	Fill in the blanks in each sentence with the correct option.	06	CO1	BTL1

i. Choose the sentence with the correct article usage:

a) I need the book to study.

b) I need book to study.

c) I need an book to study.

d) I need a book to study.

ii. Select the sentence used with correct verb tense:

a) She is going to the party yesterday.

b) She goes to the party yesterday.

c) She went to the party yesterday.

d) She go to the party yesterday.

iii. She wants to travel abroad, ____she is saving money.

a) or

b) but

c) and

d) so

iv. Choose the correct collocation:

a) They took a decision.

b) They made a decision.

c) They did a decision.

d) They had a decision.

v. Identify the sentence with the correct word order:

a) In the garden, children are playing happily.

b) In happily the garden, playing children are.

c) Happily children are playing in the garden.

d) In playing garden, children happily are.

vi. We should____early to catch the train.

a) get up

b) get off

c) get in

d) get through

2	Do as directed:	06	CO1	BTL1,3
	i. Fill in the blanks with 'a', 'an', or 'the' as necessary:			
	She wants to become ___ astronaut.			
	ii. Correct the verb tense in the sentence:			
	They has gone to the market.			
	iii. Rewrite the sentence using a suitable conjunction:			
	I will go for a walk. It stops raining.			
	iv. Correct the collocation usage in the sentence:			
	He made a decision.			
	v. Re-arrange the words to form a meaningful sentence:			
	dog / chased / cat / the / the / garden / in the/ went/ to/ park/ children.			
	vi. Choose the correct phrasal verb:			
	The meeting ___ (put off/called off) due to bad weather.			
3	Describe the communication process and the sequential flow within the communication cycle.	06	CO2	BTL2,4
Q.3	**Answer the following: (Any Two)**	**12**		
1	Discuss the relevance of different listening modes in various contexts.	06	CO3	BTL2,4
2	Illustrate the techniques of skimming and scanning in comprehending written text.	06	CO4	BTL2,6
3	Discuss the significance of possessing effective interpersonal oral conversation skills.	06	CO4	BTL2,6
Q.4	**Answer the following: (Any One)**	**06**		
1	Analyze the distinctions between formal and informal writing structures. Include examples and clarify the appropriate situations for each style.	06	CO5	BTL1,2
2	Write a job application letter to a software development company, expressing interest in the software designer position advertised. Detail your qualifications, skills, and enthusiasm for the role.	06	CO5	BTL2,4

**** End of the Paper ****